MTHS120: Mock Examination

Week 1: Sequences & Discrete Calculus

Time Allowed: 60 Minutes | **Total Marks:** 50

Instructions:

- Answer all questions in the spaces provided on this paper.
- Show all necessary algebraic working. A correct final answer without supporting mathematical justification may not receive full marks.
- For multiple-choice questions, clearly circle the letter corresponding to your chosen answer.

Part A: Definition Recall & Concepts (Easy)

This section tests your fundamental understanding of sequence properties and definitions.

1. **[4 marks]** Which of the following algebraic conditions rigorously proves that a sequence (a_n) is *strictly decreasing* for all $n \geq 1$?

   A. $a_{n+1} - a_n \leq 0$ for all $n \geq 1$.
   B. $a_{n+1} - a_n < 0$ for all $n \geq 1$.
   C. $\frac{a_{n+1}}{a_n} < 1$ for all $n \geq 1$.
   D. $a_1 > a_2 > a_3$.

2. **[4 marks]** An analysis engine evaluates a sequence (x_n) and logs that it is strictly bounded by a ceiling M (i.e., $|x_n| \leq M$ for all n). What does this guarantee about the sequence?

   A. The sequence converges to a limit $L \leq M$.
   B. The sequence is monotonic.
   C. The sequence does not escape to infinity, but it may still diverge by oscillation.
   D. By the Axiom of Completeness, the sequence must have a strictly positive limit.

Part B: Mechanical Computation (Medium)

This section requires algebraic execution of the discrete derivative and limit behavior.

3. **[6 marks]** Calculate the discrete derivative, $a'_n = a_{n+1} - a_n$, for the arithmetic sequence given by $a_n = 4n - 5$. Show your substitution clearly.

4. **[6 marks]** Compute the discrete derivative a'_n for the geometric progression $a_n = 3 \cdot (2)^n$. Factor your final answer as simply as possible.

5. **[8 marks]** Consider the rational sequence $a_n = \frac{n}{n+1}$. By evaluating its discrete derivative $a_{n+1} - a_n$, determine the monotonicity of the sequence (i.e., is it strictly increasing, strictly decreasing, or neither?).

Part C: Advanced Synthesis & Telescoping (Hard)

This section tests your ability to apply the Discrete Fundamental Theorem of Calculus to evaluate complex sums and limits.

6. **[6 marks]** Calculate the discrete derivative a'_n for the quadratic sequence $a_n = n^2 - n$.

7. **[8 marks]** Evaluate the following telescoping sum explicitly in terms of n:

$$\sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

8. **[8 marks]** Let $a_n = n^2$ and its discrete derivative be $a'_n = 2n + 1$. Using the Discrete Fundamental Theorem of Calculus (Telescoping Identity), evaluate the following sum exactly:

$$\sum_{k=1}^{n-1} (2k + 1)$$

Explain how the boundary conditions of the sum cancel the interior terms.